

Going viral models



Learning objective: To investigate different models that represent a virus.

Key skill: List characteristics of a virus.

Materials: For the paper model: scissors, templates, glue stick, sellotape (single-sided and double-sided); optional – straws, pipe cleaners, string. For the play dough model: play dough, rolling pin; optional – straws, pipe cleaners, string.

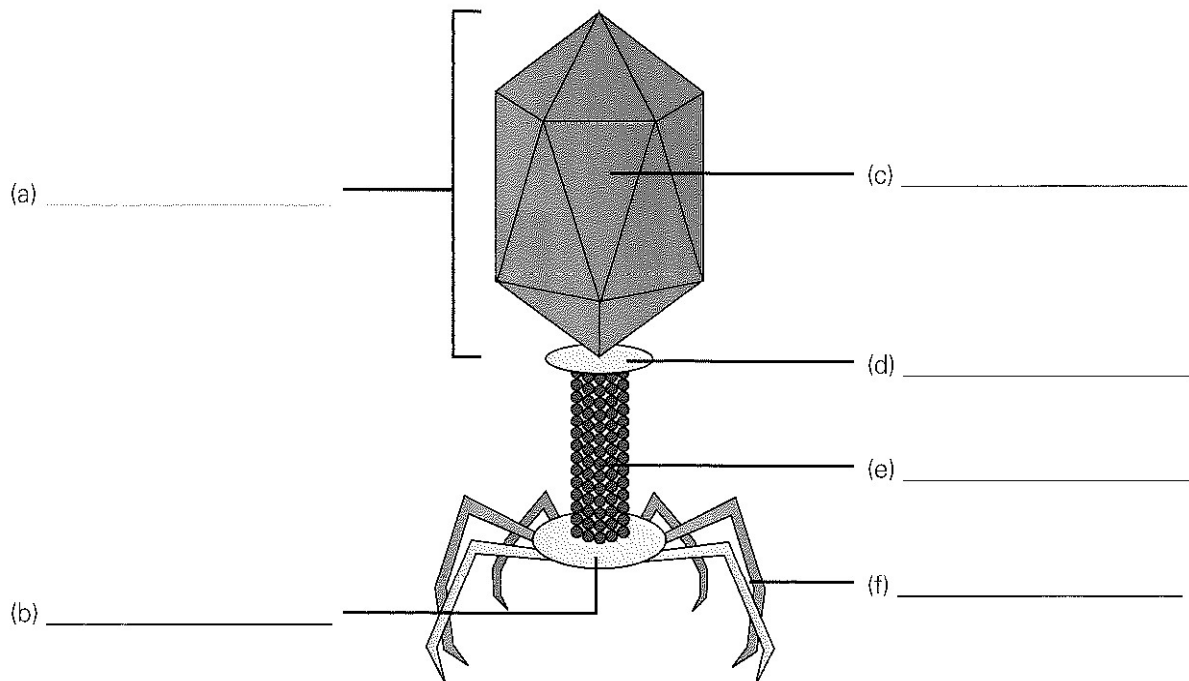
Scientific models make complicated things simple. Here you will explore four models of a virus: a diagram; images; a paper model; and a play dough model.

A virus:

- has RNA or DNA (nucleic acid)
- is a capsid
- may not be living
- is extremely tiny at 20–400 nm
- is not a cell
- has no cell nucleus
- has no cell wall
- may have a tail, collar, sheath, baseplate, fibres and spikes
- may have an external (outside) membrane.

Model 1: Diagram of a virus

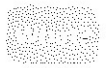
Label the diagram, using the word bank to help you.



Word bank: capsid, nucleic acid (DNA or RNA), collar, sheath, baseplate, tail fibres

continued ...

Going viral quiz



Learning objective: To identify what defines a virus.

Key skill: Complete a quiz.

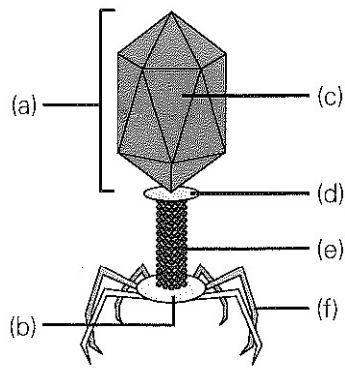
What do you know about viruses? Try this quiz twice to find out! Circle True or False after each statement.

Number correct *before* I learnt about viruses = _____

Number correct *after* I learnt about viruses = _____

- | | |
|--|--------------|
| 1. Viruses are made of cells. | True / False |
| 2. Viruses reproduce on their own. | True / False |
| 3. Viruses contain genetic material. | True / False |
| 4. Viruses can evolve. | True / False |
| 5. Viruses have an outer protein capsid and nucleic acid. | True / False |
| 6. Viruses have an outer carbohydrate capsid and nucleic acid. | True / False |
| 7. Viruses have an outer glyco-protein capsid and nucleic acid. | True / False |
| 8. Viruses may have a protein tail and carbohydrate tail fibres. | True / False |
| 9. Viruses may have a protein capsid, nucleic acid, collar, sheath, baseplate, and tail fibres and spikes. | True / False |
| 10. Viruses may have a membrane envelope with glycoproteins. | True / False |
| 11. Viruses are bacteriophages. | True / False |
| 12. Viruses are extremely tiny. | True / False |
| 13. Viruses eat or engulf bacteria. | True / False |
| 14. Viruses eat or engulf other viruses. | True / False |
| 15. Viruses are much smaller than bacteria. | True / False |
| 16. Viruses reproduce inside host cells. | True / False |

continued ...



17. On the diagram, (d) labels the nucleic acid. True / False
18. On the diagram, (d) labels the capsid. True / False
19. On the diagram, (d) labels the sheath. True / False
20. Viruses are larger than bacteria. True / False
21. Viruses may not be alive. True / False
22. Viruses have a cell wall. True / False
23. Viruses have a cell nucleus. True / False
24. Viruses can contain RNA or DNA. True / False
25. Viruses never contain RNA. True / False
26. Viruses can be killed by antibiotics. True / False
27. On the diagram, (a) labels the capsid, collar and sheath. True / False
28. Viruses may have (f) on the diagram. True / False

Viral infection

How?

Learning objective: To explore how viruses infect cells.

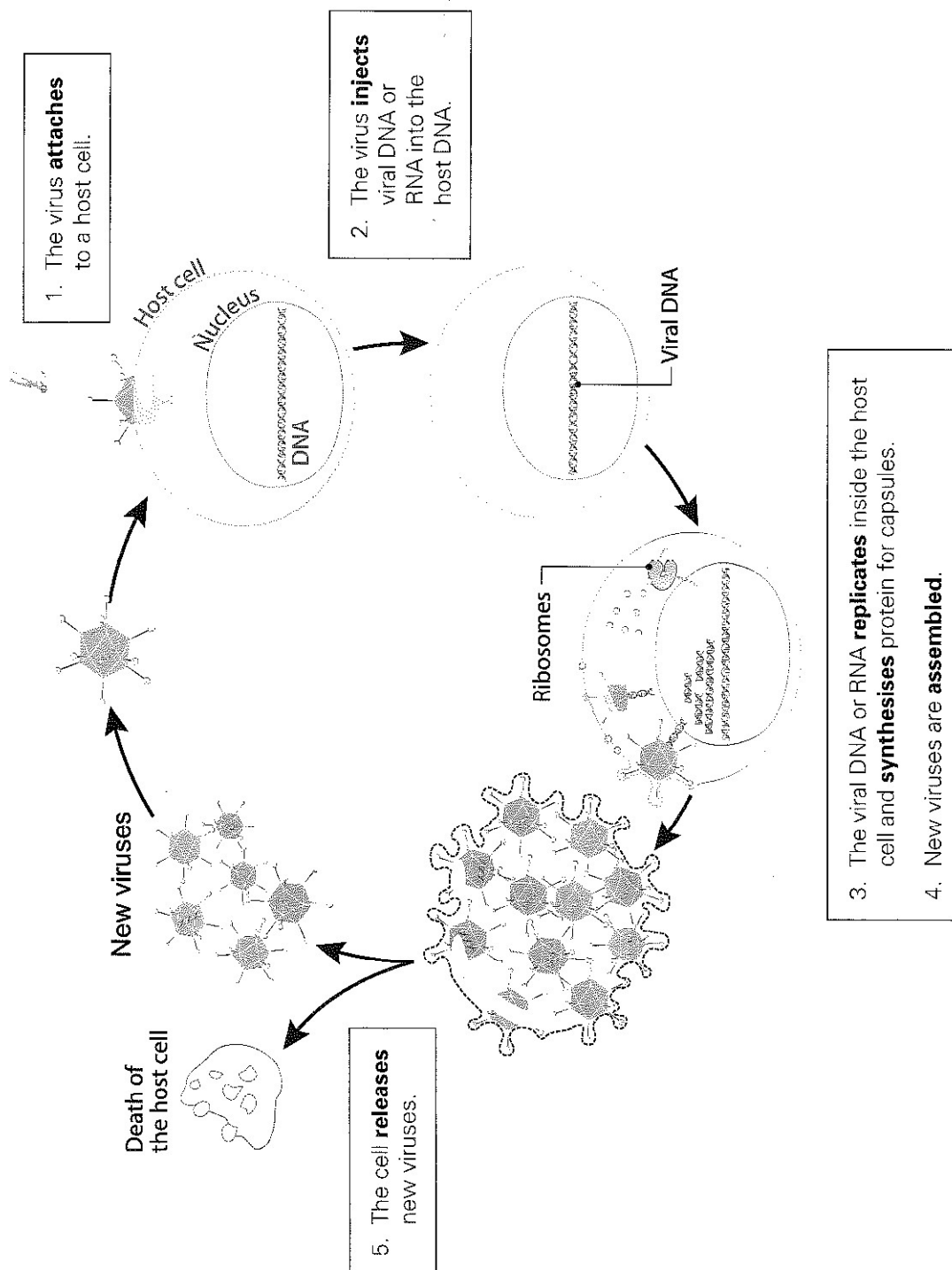
Key skills: Listen; label a diagram.

Materials: Pencil, eraser. Optional – scissors, glue stick and A4 paper for mix and match.

There are thousands of different kinds of viruses. Viruses cannot be killed by antibiotics.

Bacteriophages are viruses that “eat” bacteria. They are found in the mucus of your mouth, nose and intestine.

LIFE CYCLE OF VIRUSES



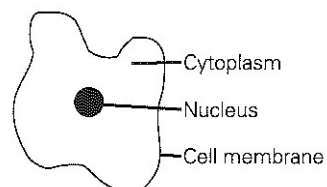
continued ...

Viral infection picture dictation

Your teacher will dictate (read out) a stage of viral infection and a description of the stage twice. Listen the first time, with your pencil on your desk. The second time, draw a diagram in pencil of each of the stages in the table.

Stage	Diagram
Attach	
Inject	
Replicate and synthesise	
Assemble	
Release	

Hint: Use these shapes or similar in your diagrams.



Virus capsule



DNA or RNA

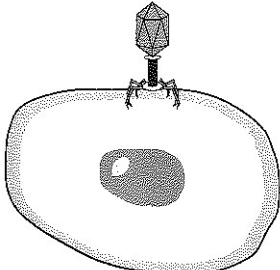
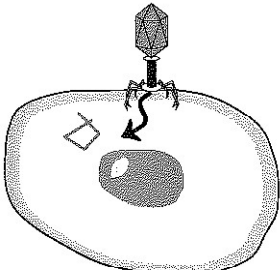
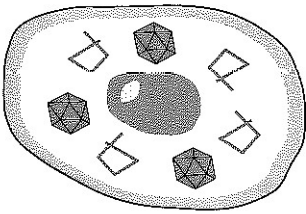
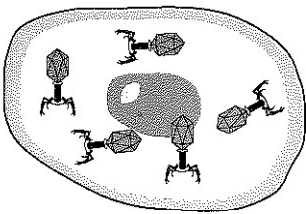
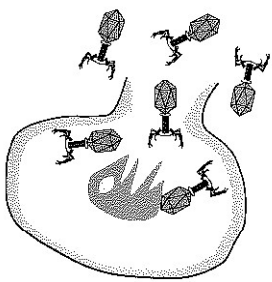
Check

- ☐ In pencil
- ☐ Ruled label lines
- ☐ Labels
- ☐ Continuous lines
- ☐ No shading

continued ...

Mix and match (optional activity)

Cut out the cards, mix them up and then match them.

Stage	Description	Diagram
Attach	The virus attaches to the host cell surface.	 A diagram showing a virus with a hexagonal head and tail fibers attaching to the surface of a host cell. The host cell is an oval shape with a nucleus inside.
Inject	The virus injects nucleic acid (DNA or RNA) into the host cell.	 A diagram showing a virus with its tail fibers embedded in the host cell membrane, injecting its nucleic acid (represented by a small square) into the host cell's nucleus.
Replicate and synthesise	The virus nucleic acid replicates and makes protein for capsules.	 A diagram showing a host cell with several virus nucleic acids (small squares) and protein capsids (hexagons) inside, indicating replication and synthesis.
Assemble	The virus nucleic acid assembles new viruses.	 A diagram showing a host cell with several assembled viruses (hexagonal heads with tail fibers) inside, indicating assembly.
Release	The virus nucleic acid releases viruses.	 A diagram showing a host cell with several assembled viruses inside, and some viruses are shown outside the cell, indicating release.